

Inductor Selection

For most applications, the value of the inductor will fall in the range of $1\mu\text{H}$ to $4.7\mu\text{H}$. Its value is chosen based on the desired ripple current. Large value inductors lower ripple current and small value inductors result in higher ripple currents. Higher V_{IN} or V_{OUT} also increases the ripple current as shown in Equation 1. A reasonable starting point for setting ripple current is $\Delta I_L = 400\text{mA}$ (40% of 1A).

$$\Delta I_L = \frac{1}{f \times L} V_{\text{OUT}} \left(1 - \frac{V_{\text{OUT}}}{V_{\text{IN}}} \right) \quad \text{Equation (1)}$$

The DC current rating of the inductor should be at least equal to the maximum load current plus half the ripple current to prevent core saturation. Thus, a 1.4A rated inductor should be enough for most applications ($1\text{A} + 400\text{mA}$). For better efficiency, choose a low DC-resistance inductor.

V_O	1.2V	1.5V	1.8V	2.5V	3.3V
L	2.2 μH	2.2 μH	2.2 μH	4.7 μH	4.7 μH

Fig. 3: Datasheet screenshot table for selection of inductor

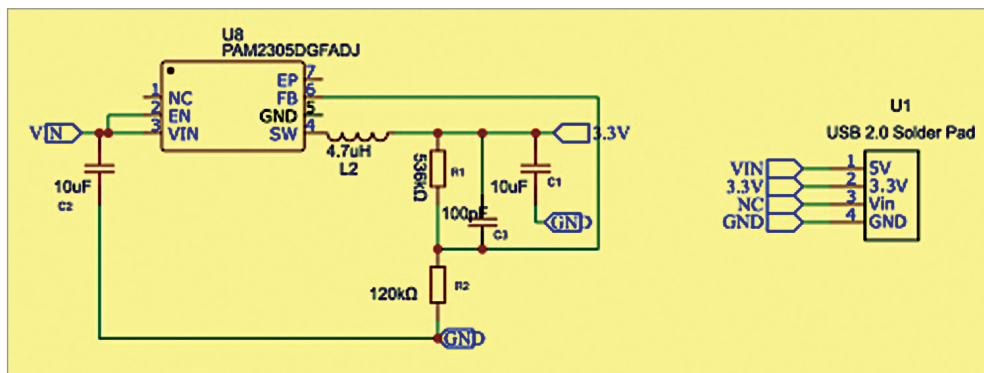


Fig. 4: Buck converter circuit for fixed voltage

pin controls the IC's operation when tied to a pull-up resistor or directly to the input voltage, enabling the IC to be active. When pulled low, it disables the converter. The SW (Switch) pin connects to an inductor, and the output voltage is set using two resistors configured between the SW, FB, and GND pins. Fig. 2 shows

the IC (PAM2305DGFADJ) datasheet pinout and application circuit.

Selecting an inductor

As stated in the inductor selection table of the datasheet, the inductor value can range from $1\mu\text{H}$ to $4.7\mu\text{H}$. For an output voltage range of 2.5V to 3.3V, a $4.7\mu\text{H}$ inductor is recommended to ensure stability and efficiency, in line with the provided guidelines. Since the complete circuit needs to fit within a 9mm area, we have selected an SMD version of the inductor with an L0603 footprint. Fig. 3 shows the datasheet screenshot table for choosing the inductor.

Selecting resistors

To configure the output voltage of the PAM2305DGFADJ buck converter, refer to the formula provided in the datasheet (see Fig. 3

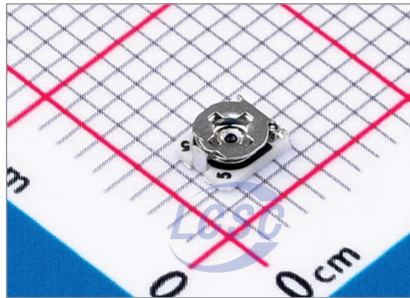


Fig. 5: SMD potentiometer used in variable circuit

compact DFN-6 ($2 \times 2\text{mm}$) package. Fig. 1 shows the author's DC-DC buck converter integrated into the custom-designed PCB.

According to the datasheet of the IC selected for this converter design (for details, refer to the IC datasheet), the design requires an inductor, several resistors, and capacitors. If a variable output voltage is desired, a potentiometer can be used. The complete

V_O	R1	R2
1.2V	100k	100k
1.5V	150k	100k
1.8V	200k	100k
2.5V	380k	120k
3.3V	540k	120k

list of needed SMD components is provided in Table 1.

Design

The buck converter IC (PAM-2305DGFADJ) comes with six pins. According to the application circuit in the datasheet, it includes a VIN pin for the input voltage, a GND pin for grounding, and an FB pin that receives feedback to regulate the output voltage. The EN (Enable)